Indian Institute of Information Technology Vadodara
Mid-Semester Examination, Autumn 2025-26
CS443: System Administration and Maintenance

Exam Duration: 2 Hours

Max. Marks: 30

General Instructions:

- **Questions must be answered in the given order, else a penalty will be awarded.**
- Use clear and precise reasoning. Marks are awarded for clarity and correctness.
- Do not write anything, except your name and id on the question paper.

1a: Explain operating systems functions/services. Define Process, PCB and attributes of a process. Draw and explain significance of each state in the process state diagram. **[5 Marks]**

10

1b: Explain SJF burst time prediction techniques in detail. Explain HRRN scheduling algorithm. What are the roles and responsibilities of a system administrator. **[5 Marks]**

2a: Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
AT	0	1	3	4
BT	3	1	3	Z

These processes are run on a single processor using the preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is? **[5 Marks]**

5

2b: Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order *P*1, *P*2, *P*3, *P*4.

Processes	BT (in ms)
P1	8
P2	7
P3	2
P4	4

WT = TAT - BT
RT = CPU - AT
TAT = CT - AT

10.5

15.75

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is? **[5 Marks]**

3

3a: Explain the need for synchronisation with example and conditions/requirements for synchronisation mechanisms. Explain sleep and wake system calls. **[5 Marks]**

4

3b: Describe the key concept of Peterson's algorithm for process synchronization in detail. Compare UNIX and Linux OS. **[5 Marks]**